

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
Student Name	Samyuktha V
Register Number	3122247001053
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	12/08/26

1 Objective

To implement linear and regularized regression models (Linear, Ridge, Lasso, Elastic Net) for predicting a continuous target variable (loan amount requested), evaluate their performance using multiple regression metrics, visualize model behavior, and analyze overfitting, underfitting, and bias-variance characteristics.

2 Problem Statement

Given a set of numerical and categorical features describing a loan applicant (income, credit score, property details, employment information, etc.), predict the **loan amount requested (USD)** as a continuous value. This is a supervised regression problem: the input is the applicant/property feature vector, and the output is a single real-valued prediction.

3 Dataset Description

Dataset Name	Loan Amount Prediction Dataset
Dataset Source	Kaggle – Predict Loan Amount Data
Number of Samples	20,000
Number of Features	19 raw features → 43 after one-hot encoding (identifier columns Customer ID, Name, Property ID were dropped)
Number of Classes	Not applicable – regression task, target is continuous (Loan Amount Request in USD)
Missing Values	Present in Gender, Income, Income Stability, Type of Employment, Current Loan Expenses, Dependents, Credit Score, Has Active Credit Card, Property Age, Property Location. Additionally, Co-Applicant and Property Price used the placeholder "?" instead of a blank cell (77 and 168 rows respectively), which was converted to NaN before imputation. After cleaning: 0 missing values.
Train-Test Split	80% train (16,000 rows) / 20% test (4,000 rows) – <code>test_size=0.20, random_state=42</code>

4 Exploratory Data Analysis

- Dataset overview (`df.head()`, `df.info()`)
- Statistical summary
- Missing value analysis (before/after cleaning, including the "?" placeholder fix)
- Target variable distribution (histogram of Loan Amount Request)
- Feature-vs-target scatter plots
- Feature distribution

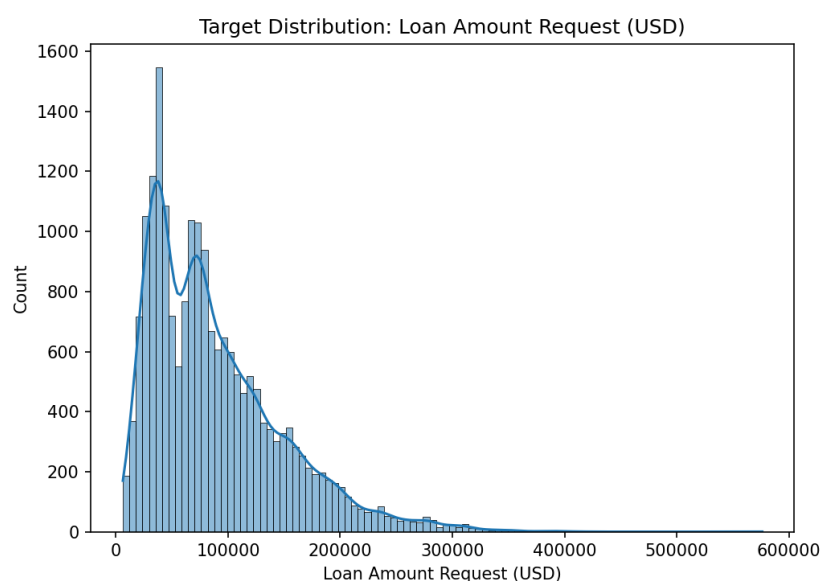


Figure 1: Distribution of the target variable: Loan Amount Request (USD)

Inference:

The target variable is right-skewed, with most loan requests concentrated in the lower range and a long tail of larger requested amounts. This means a small number of high-value applications pull the mean above the median. Because ordinary least squares regression minimizes squared error, these large-value tail cases have an outsized influence on the fitted line – one reason regularized variants (Ridge, Lasso, Elastic Net) are worth comparing against plain Linear Regression here.

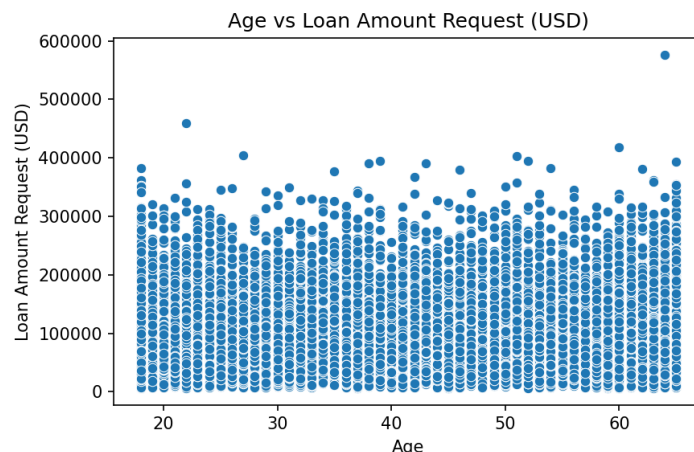


Figure 2: Age vs. Loan Amount Request (USD)

Inference:

The scatter plot of Age against the requested loan amount shows a wide, largely formless spread with no strong visible linear trend – applicants across most age groups request a broad range of loan amounts. This suggests Age alone is a weak individual predictor, and the model needs to rely on combinations of several features (income, credit score, property price, etc.) rather than any single feature to explain the target well.

5 Data Preprocessing

- **Missing value handling:** placeholder strings ("?") were first converted to real NaN values, since two columns (**Co-Applicant**, **Property Price**) used "?" instead of a blank cell and were being silently misread as text. Numeric columns were then imputed with the column median (robust to outliers); categorical columns were imputed with the column mode.
- **Encoding:** categorical columns were one-hot encoded using `pd.get_dummies()` with `drop_first=True` to avoid the dummy variable trap. This expanded the feature set from 19 to 43 columns.
- **Feature scaling:** all numeric features were standardized using `StandardScaler` (mean = 0, standard deviation = 1), fit on the training set only and applied to both train and test to avoid data leakage.
- **Train-test split:** 80/20 split (16,000 / 4,000 rows), no stratification (stratification does not apply to a continuous regression target).
- **Feature engineering:** identifier columns (Customer ID, Name, Property ID) were dropped as they carry no predictive signal.

6 Machine Learning Algorithm

Linear Regression

Working principle: fits a hyperplane that minimizes the sum of squared residuals between predicted and actual values.

Mathematical formulation: $\hat{y} = w_0 + \sum_{i=1}^n w_i x_i$, minimizing $\sum (y - \hat{y})^2$.

Advantages: simple, interpretable, fast to train.

Limitations: prone to overfitting with many/correlated features; sensitive to outliers.

Ridge Regression (L2 Regularization)

Working principle: adds a penalty proportional to the square of the coefficients to the loss function, shrinking coefficients toward zero.

Mathematical formulation: minimizes $\sum (y - \hat{y})^2 + \alpha \sum w_i^2$.

Advantages: reduces overfitting, handles multicollinearity well.

Limitations: coefficients rarely become exactly zero, so it does not perform feature selection.

Lasso Regression (L1 Regularization)

Working principle: adds a penalty proportional to the absolute value of the coefficients.

Mathematical formulation: minimizes $\sum (y - \hat{y})^2 + \alpha \sum |w_i|$.

Advantages: performs automatic feature selection by driving some coefficients to exactly zero.

Limitations: can be unstable when features are highly correlated (arbitrarily picks one).

Elastic Net Regression

Working principle: combines the L1 and L2 penalties above, controlled by a mixing ratio.

Mathematical formulation: minimizes $\sum (y - \hat{y})^2 + \alpha (l1_ratio \sum |w_i| + (1 - l1_ratio) \sum w_i^2)$.

Advantages: balances Ridge's stability with Lasso's feature selection.

Limitations: two hyperparameters to tune instead of one, more expensive to search.

7 Experimental Setup

Python Version	3.11
Scikit-Learn Version	1.9.0
IDE	Jupyter Notebook
Operating System	Windows
Hardware	Intel i5, 16GB RAM

8 Hyperparameter Selection

Parameter	Search Space
Ridge α	0.01, 0.1, 1, 10, 100
Lasso α	0.001, 0.01, 0.1, 1, 10
Elastic Net α	0.01, 0.1, 1, 10
Elastic Net $l1_ratio$	0.2, 0.5, 0.8
Cross-Validation Folds	5
Search Methods	GridSearchCV, RandomizedSearchCV

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	GridSearchCV	$\alpha = 10$	0.9302
Ridge Regression	RandomizedSearchCV	$\alpha = 10$	0.9302
Lasso Regression	GridSearchCV	$\alpha = 10$	0.9302
Lasso Regression	RandomizedSearchCV	$\alpha = 10$	0.9302
Elastic Net Regression	GridSearchCV	$\alpha = 0.01, l1_ratio = 0.8$	0.9302
Elastic Net Regression	RandomizedSearchCV	$\alpha = 0.01, l1_ratio = 0.8$	0.9302

9 Results

5-Fold Cross-Validation Performance (Training Set)

Model	MAE	MSE	RMSE	R^2
Linear Regression	11,092.82	2.507×10^8	15,833.26	0.9302
Ridge Regression	11,092.80	2.507×10^8	15,833.22	0.9302
Lasso Regression	11,092.06	2.507×10^8	15,832.77	0.9302
Elastic Net Regression	16,967.75	5.211×10^8	22,827.91	0.8550

Test Set Performance (Tuned Models)

Model	MAE	MSE	RMSE	R^2
Linear Regression	10,898.80	2.331×10^8	15,268.73	0.9355
Ridge Regression	10,899.42	2.331×10^8	15,269.07	0.9355
Lasso Regression	10,896.69	2.330×10^8	15,263.43	0.9356
Elastic Net Regression	10,900.97	2.332×10^8	15,270.28	0.9355

Note: MAE/MSE/RMSE/ R^2 are reported instead of Accuracy/Precision/Recall/ F1/ROC-AUC, since those classification metrics do not apply to a continuous-target regression problem.

10 Result Visualization

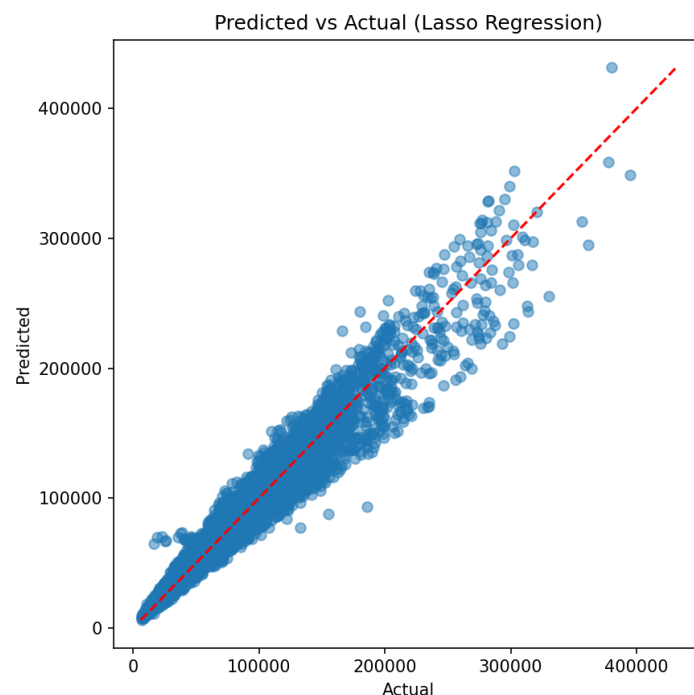


Figure 3: Predicted vs. Actual Loan Amount Request (Lasso Regression, best tuned model)

Inference:

Points cluster tightly around the diagonal red dashed line for most of the range, indicating the model predicts the majority of loan amounts accurately. Spread increases visibly at higher actual loan values, showing the model is somewhat less precise for large, less common loan requests – consistent with the right-skewed target distribution seen earlier.

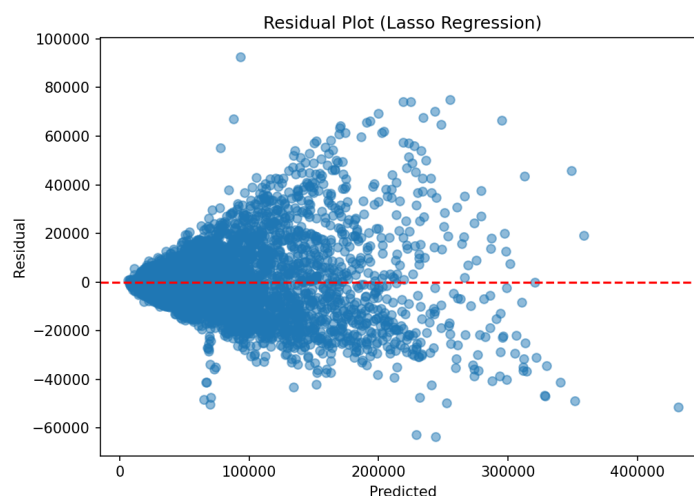


Figure 4: Residual Plot (Lasso Regression, best tuned model)

Inference:

Residuals are centered around zero across most of the predicted range with no strong curved pattern, suggesting the linear assumption is reasonable for this data. Residual spread widens somewhat at higher predicted values (a mild "funnel" shape), indicating slightly higher variance in error for larger loan predictions – a case of mild heteroscedasticity rather than a systematic model bias.

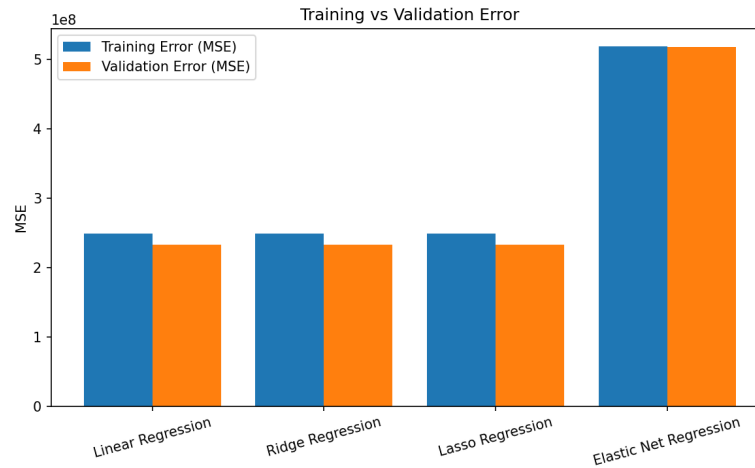


Figure 5: Training vs. Validation Error (MSE) across all four baseline models

Inference:

Linear, Ridge, and Lasso Regression show training and validation error very close to each other, indicating a good fit with no significant overfitting. Elastic Net's default (untuned) parameters produced a noticeably higher error on both train and validation sets, showing it was underfitting before tuning – explaining why its baseline R^2 (0.857) trailed the other three models until hyperparameter search corrected it (tuned $R^2 = 0.9355$).

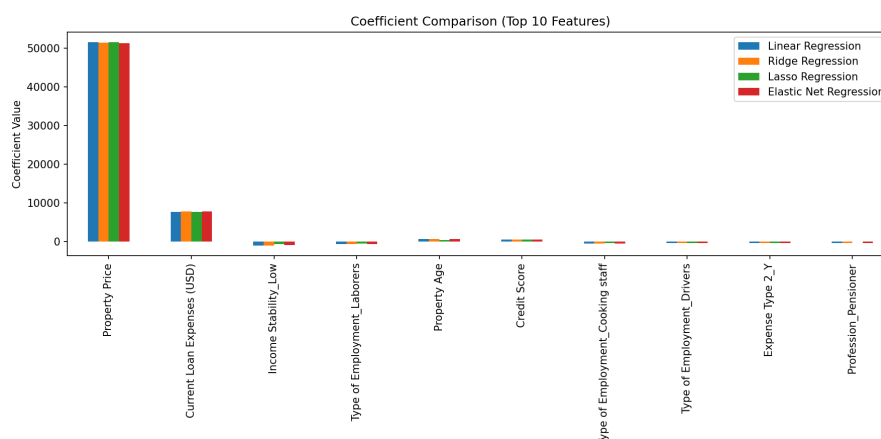


Figure 6: Coefficient Comparison across tuned models (top 10 features by magnitude)

Inference:

The four models assign broadly similar signs and relative magnitudes to the top features, which is expected given how close their α values ended up after tuning. None of

Lasso's coefficients were driven exactly to zero in this run (0 of 43), suggesting most one-hot-encoded and numeric features carry at least some predictive signal for this dataset rather than being pure noise.

11 Discussion

Overfitting and Underfitting Analysis

Training and validation error are very close for Linear, Ridge, and Lasso Regression, indicating none of them overfit noticeably on this dataset – with 20,000 rows and only 43 features after encoding, there is enough data relative to model complexity that regularization changes very little. Elastic Net's default parameters (before tuning) underfit, evidenced by higher error on *both* train and validation sets; tuning found a small $\alpha = 0.01$ with $l1_ratio = 0.8$ that brought its performance in line with the other three models.

Bias–Variance Analysis

Linear Regression shows low bias (it fits the training data well) and, on this dataset, low variance too, since regularization barely changed its coefficients or test performance. Ridge's shrinkage had a negligible effect here (best $\alpha = 10$ still yielded near-identical R^2 to plain Linear Regression), suggesting the original model was not suffering from high variance/multicollinearity in the first place. Lasso, despite selecting $\alpha = 10$, retained all 43 coefficients as non-zero – indicating limited feature sparsity was needed, i.e. most encoded features contribute at least marginally to prediction.

12 Conclusion

Lasso Regression ($\alpha = 10$) gave the best test-set performance ($R^2 = 0.9356$, RMSE = 15,263.43), marginally ahead of Linear, Ridge, and the tuned Elastic Net – all four converge to nearly identical performance once tuned, which suggests the relationship between the features and loan amount is close to linear and well-conditioned, leaving little room for regularization to improve on plain Linear Regression. The main practical value of tuning in this experiment was correcting Elastic Net's underfitting default configuration rather than improving an already-strong baseline.

13 References

1. Scikit-learn: Linear Models, https://scikit-learn.org/stable/modules/linear_model.html
2. Scikit-learn: Hyperparameter Optimization, https://scikit-learn.org/stable/modules/grid_search.html
3. Kaggle: Predict Loan Amount Data